

***Nut consumption is associated with decreased health risk factors
for cardiovascular disease and metabolic syndrome in U.S. adults:
NHANES 1999-2004.***

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BACKGROUND: Few recent epidemiologic studies have assessed the effect that nut consumption (including tree nuts and peanuts) has on health risks, including metabolic syndrome (MetS). OBJECTIVE: This study compared the health risk for cardiovascular disease, type 2 diabetes, and MetS of nut consumers with that of nonconsumers. DESIGN: Adults 19+ years (n = 13,292) participating in the 1999-2004 National Health and Nutrition Examination Survey were used. Intake from 24-hour recalls was used to determine intake. Nut/tree nut consumers consumed $\geq \frac{1}{4}$ ounce per day. Covariate-adjusted means, standard errors, and prevalence rates were determined for the nut consumption groups. RESULTS: The prevalence of nut consumers was $18.6\% \pm 0.7\%$ and $21.0\% \pm 0.9\%$ in those 19-50 years and 51 years and older, respectively. Nut consumption was associated with a decreased body mass index ($27.7 \text{ kg/m}^2 \pm 0.2$ vs $28.1 \pm 0.1 \text{ kg/m}^2$, $p < 0.05$), waist circumference ($95.6 \pm 0.4 \text{ cm}$ vs $96.4 \pm 0.3 \text{ cm}$, $p < 0.05$), and systolic blood pressure ($121.9 \pm 0.4 \text{ mmHg}$ vs $123.20 \pm 0.3 \text{ mmHg}$, $p < 0.01$) compared with nonconsumers. Tree nut consumers also had a lower weight ($78.8 \pm 0.7 \text{ kg}$ vs $80.7 \pm 0.3 \text{ kg}$, $p < 0.05$). Nut consumers had a lower percentage of two risk factors for MetS: hypertension ($31.5\% \pm 1.0\%$ vs $34.2\% \pm 0.8\%$, $p < 0.05$) and low high density lipoprotein-cholesterol (HDL-C) ($29.6\% \pm 1.0\%$ vs $34.8\% \pm 0.8\%$, $p < 0.01$). Tree nut consumers had a lower prevalence of four risk factors for MetS: abdominal obesity ($43.6\% \pm 1.6\%$ vs $49.5\% \pm 0.8\%$, $p < 0.05$), hypertension ($31.4\% \pm 1.2\%$ vs $33.9\% \pm 0.8\%$, $p < 0.05$), low HDL-C ($27.9\% \pm 1.7\%$ vs $34.5\% \pm 0.8\%$, $p < 0.01$), high fasting glucose ($11.4\% \pm 1.4\%$ vs $15.0\% \pm 0.7\%$, $p < 0.05$), and a lower prevalence of MetS ($21.2\% \pm 2.1\%$ vs $26.6\% \pm 0.7\%$, $p < 0.05$). CONCLUSION: Nut/tree nut consumption was associated with a decreased prevalence of selected risk factors for cardiovascular disease, type 2 diabetes, and MetS.